

The Data Analytics Journey

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Ghavami (2020) defines data analytics as "a set of methods, processes, and disciplines that extract, analyze, and transform data into meaningful insight, discovery, and knowledge to make more effective decisions" (p.4). Though experiences with data analytics may vary from person to person, some foundational elements remain constant. The analytics life cycle, the tools and techniques used, and evaluating legal and ethical risks all play a role in an analyst's job. This paper explores these core components to understand how data analysis functions in different industries.

The Phases of the Data Analytics Life Cycle

The Data Analytics Life Cycle consists of seven phases that guide the transformation of raw data into a reliable outcome. Though the cycle structure provides a standardized framework, it is not linear, and an analyst may not always use every phase in a project. Analysts must understand each phase to ensure the results are effective and the data is accurate.

Business Understanding

Business understanding is the first phase in the Data Analytics Life Cycle, also known as the discovery/planning phase. An analyst gains expertise during this stage by engaging with stakeholders to understand the organization's goals, mission, and needs. For example, in the healthcare staffing industry, the mission may be to ensure the timely placement of qualified clinicians in open roles. This mission helps the analyst identify key business requirements, such as tracking the number of open positions and understanding staffing trends. The analyst can define the project's scope and align it with organizational priorities by asking stakeholders how frequently the analysis is needed and what data sources are available.

Data Acquisition

Data acquisition is the second phase in the Data Analytics Life Cycle. Analysts extract data from various sources such as SQL databases, Power BI dashboards, or cloud platforms in this phase. Analysts gain expertise by learning to navigate and query these data sources effectively. For example, in the beauty industry, Sephora may use SQL to hold consumer purchasing data. If one of the organization's goals is to enhance customer satisfaction through personalized product recommendations (i.e., showing different lip liners if a customer is viewing lipstick), the analyst must understand how to retrieve relevant data from SQL, such as purchase history and what the customer shows interest in. The positioning between the organization's mission and the data helps the analyst to support the business's strategic goals.

Data Cleaning

Data Cleaning is the third phase of the Data Analytics Life Cycle. Also known as data wrangling, this phase involves modifying and transforming raw data to improve its quality. Analysts develop their skills using tools like SQL, Python, R, or Excel to detect and correct errors, handle missing values, and standardize formats. For example, Pfizer may collect clinical trial data from hospitals and patients. If the organization's mission is to develop safe and effective medications, the analyst must ensure the data is accurate and consistent. Verify things like patient demographics, align dosage records, and remove duplicates. This phase is vital because inconsistencies can lead to misleading or inaccurate data.

Data Exploration

Data exploration is the fourth phase in the Data Analytics Life Cycle. During this phase, analysts learn the primary nature of the data and the relationships between variables. This phase uses data visualization tools, central tendency, and variability. Analysts can gain expertise by

interpreting visualizations and statistical summaries to uncover patterns and trends. Stakeholders for PepsiCo, Inc. may want an analyst to explore sales data across different regions during a specific season (i.e, spring or summer) to identify consumer buying trends. Suppose the organization's mission is to increase the number of products tailored to customer preferences during the warmer months. In that case, the analyst uses this phase to detect relationships, such as increased demand for citrus-flavored drinks during the summer, that steer future product development. By aligning exploration with the company's objectives, analysts can ensure that the result is relevant and effective.

Predictive Modeling

Predictive Modeling is the fifth phase in the cycle. During this phase, tools like Python and R are used to build and clean algorithms efficiently. Analysts gain expertise by selecting appropriate modeling techniques, such as regression and clustering, and using them to improve accuracy and reliability. For instance, an analyst with Chevrolet might develop a model to predict vehicle maintenance needs based on driving behavior and sensor data. If one of the organization's missions is to enhance car durability and customer satisfaction, predictive modeling helps identify which vehicles will likely require service soon. By enabling proactive outreach and scheduling, analysts ensure that predictions translate into strategic advances by aligning modeling efforts with the organization's goals.

Data Mining

Data mining is the sixth phase in the Data Analytics Life Cycle. In this phase, the analyst applies advanced techniques, using tools such as Python, to discover hidden patterns and irregularities within datasets. During this phase, analysts gain expertise by mastering these tools and techniques to discover hidden trends. For example, an analyst for Doritos may use data

mining to discover that consumers who buy Spicy-Nacho chips frequently purchase Cool Ranch as well. If one of the organization's goals is to generate more revenue, these insights can help stakeholders with decisions on bundling strategies, target promotions, or even shelf placement decisions. Analysts transform complex datasets into strategic opportunities by mining data with business goals in mind.

Reporting and Visualization

Reporting and visualization are the final phase in the Data Analytics Life Cycle. During this phase, the analyst communicates findings through clear, compelling visuals and reports that translate complex analysis into clean and concise data. Tableau and Power BI are both used to create interactive dashboards and charts. Analysts strengthen their expertise in visualization by learning how to adjust to different audiences, ensuring that insights are straight to the point yet still impactful. An analyst for Prada might visualize seasonal trends across product categories to help designers and merchandisers plan upcoming collections. If one of the organization's missions is to stay ahead of consumer preferences and deliver trend-driven products, effective reporting helps decision-makers act quickly and confidently. By aligning visual storytelling with business goals, analysts turn data into influence.

Data Analytics Tools and Techniques

Decision-Making Process

An analyst in the housing industry would know that selecting the appropriate data analytics technique during the predictive modeling phase of the Data Analytics Life Cycle is essential for accurate forecasting and decision-making. Regression analysis is better than alternatives such as classification and clustering because it can model relationships between

multiple predictor variables, such as location, square footage, and market trends, and a continuous variable, like housing price. For example, an organization may face challenges if the data from traditional appraisal methods fail to reflect real-time market pricing. This could result in confusion among potential buyers and missed sale opportunities. Analysts can build a dynamic pricing model that adapts to new data and provides reliable, data-driven estimates by implementing regression techniques into the predictive modeling phase. Shukla (2024) demonstrates that multiple linear regression effectively predicts housing prices by analyzing key variables and evaluating model performance using metrics like R-squared and RMSE, confirming its robustness in real estate forecasting. The regression technique in the predictive modeling phase can strengthen consumer confidence and support strategic decision-making in a competitive housing market.

Legal and Ethical Risks

Risks using the Regression Technique

Floridi and Taddeo (2016) define data ethics as a branch concerned with the moral challenges of data, algorithms, and digital practices. Davison et al. (2021) define data privacy as protecting stored or collected data through technical safeguards like encryption and access control. Using the regression technique during the data analytics life cycle, particularly in the predictive modeling phase, comes with several legal and ethical risks.

The regression technique relies on large datasets that may contain sensitive consumer or business data. An example of sensitive information is patient records. Healthcare organizations could face legal penalties and reputational damage if laws such as the Health Insurance Portability and Accountability Act (HIPAA) are violated.

Analysts should also be cautious about the wrong use of data when handling it. Predictive modeling can reveal sensitive information such as the income level of consumers. Even if not intended to be collected, misuse of this data in financial organizations could raise concerns about discriminatory outcomes.

Lastly, bias and discrimination can surface if raw data is skewed. For example, if an analyst in the candy industry uses the regression technique and the data reflects biased assumptions about high sugar intake in specific neighborhoods, the model could reinforce harmful stereotypes.

These risks pose ethical challenges and threaten essential business functions such as compliance and a company's values and ethics.

Conclusion

When conducting analysis, it is essential to consider the phases of the data analytics life cycle.

While the steps may vary from project to project, a clear understanding of each phase is essential to deliver adequate results. Analysts should remain mindful of the tools and techniques available throughout the process, making sure to select those that best support the needs of the type of analysis they are conducting. Lastly, legal and ethical risks should remain a top priority. Being aware of risks helps to prevent bias and discrimination, and the commitment to responsible data practices.

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